
When Does Grounding Help LLM Ideation? A Single-Protocol Ablation of Six Grounding Mechanisms

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Abstract

A popular prescription for automated scientific ideation is to *ground* the process—in deduction, induction, small experiments, literature comparison, or proposal writing—rather than prompting a language model abstractly. Yet every prior method grounds in only one way and compares against a (typically weak) ungrounded baseline, so it remains unknown which grounding *mechanism* actually helps when all else is held fixed. We present the first single-protocol, head-to-head ablation of six ideation regimes (one ungrounded control plus five grounding mechanisms). Holding the model (GPT-4.1-MINI), task suite, generation budget, and evaluation procedure fixed, each regime produces a bank of 15 task hypotheses, and each bank is scored by an objective downstream-utility metric: how accurately an LLM applying the bank predicts held-out labels, both in-distribution (IND) and out-of-distribution (OOD), across three classification tasks and three seeds (54 banks, $\approx 11.9\text{k}$ API calls). We find that grounding does *not* uniformly help: pooled across regimes, grounded banks are statistically indistinguishable from—and slightly below—the ungrounded control. But the mechanism matters decisively. Grounding in *empirical feedback* (test-and-revise) is the only regime that significantly improves IND accuracy (+4.4 points, Cohen’s $d = 1.09$, $p = 0.02$), at roughly $80\times$ the token cost; proposal writing significantly *hurts* (−5.7 points, $p = 0.008$) and collapses idea diversity; and data grounding can *overfit*, trading OOD robustness for IND fit. Novelty and diversity do not track usefulness within a task. For today’s strong LLMs, “ground your ideation” is too coarse: only empirical grounding reliably adds value, and ungrounded prompting is a surprisingly hard baseline to beat.

1 Introduction

Can large language models (LLMs) generate *good* scientific ideas? Skepticism is widespread: model-generated hypotheses tend to be either incremental or implausibly radical, and “novelty” itself resists definition [Si et al., 2025]. A recurring answer in recent work is to *ground* ideation rather than prompt for ideas abstractly. SciMON grounds in retrieved literature [Wang et al., 2024]; HypoGeniC grounds in labeled data [Zhou et al., 2024]; ResearchAgent grounds in structured proposals [Baek et al., 2025]; and SGA grounds in simulation feedback [Ma et al., 2024]. Each reports that its grounded method beats an ungrounded prompt.

Why this matters. If we can say *which kinds of grounding* turn raw LLM ideation into useful ideas—and at what cost—we give practitioners a concrete recipe and move the field past single-number “novelty” leaderboards toward grounded usefulness. This is increasingly urgent because the ungrounded baseline is a moving target: the 2023-era finding that grounding helps was measured against much weaker models, and a strong modern LLM may already encode much of what grounding was meant to supply.

The gap. Every prior method grounds in *one* way and compares to an ungrounded prompt. Nobody has held task, model, generation budget, and evaluation metric fixed and ablated the grounding *mechanisms* against each other. So we cannot say whether literature comparison, induction, empirical feedback, deduction, or proposal writing is responsible for the reported gains—or whether, with a strong 2025 model, any of them still helps. We also heed the field’s central caveat, that novelty is over-trusted and can flip once ideas are executed [Si et al., 2025], by scoring *grounded usefulness* rather than a bare novelty score.

Our approach. We build the first single-protocol, head-to-head ablation of six ideation regimes: one ungrounded control (C0) and five grounding mechanisms (deduction, induction, empirical small-experiments/outcome-imagination, literature comparison, and proposal writing). We operationalize an “idea” as a *task hypothesis*—an explicit decision rule explaining a binary classification phenomenon—and “better” as how accurately an LLM applying a bank of 15 such hypotheses predicts held-out labels, in-distribution (IND) and out-of-distribution (OOD). Holding the model (GPT-4.1-MINI), bank size, and an identical evaluator fixed, the *only* variable is how the ideas were grounded. We run all six regimes on three HYPOGENIC tasks across three seeds (54 banks, ≈ 11.9 k API calls, $\approx \$3$), and complement accuracy with diversity, a novelty–usefulness analysis, and a cost/benefit accounting.

What we find. Grounding does not uniformly help. Pooled across regimes, grounded banks are statistically indistinguishable from—and slightly below—the ungrounded control (IND -1.2 pts, OOD -2.5 pts, both n.s.). But the mechanism matters decisively: empirical small-experiment grounding is the only regime that significantly improves IND accuracy ($+4.4$ pts, Cohen’s $d = 1.09$, $p = 0.02$), while proposal writing significantly *hurts* (-5.7 pts, $d = -1.56$, $p = 0.008$) and collapses diversity. Data grounding can *overfit*: on deceptive reviews the ungrounded model transfers near-perfectly OOD (0.97) using general priors, whereas data-grounded rules latch onto dataset-specific surface cues and drop to 0.78–0.82 OOD. Empirical grounding’s gains cost roughly $80\times$ the generation tokens of the control.

Contributions.

- We conduct the first single-protocol, head-to-head ablation of six LLM ideation regimes, holding model, task suite, generation budget, and evaluation fixed and scoring each by objective downstream utility (IND and OOD accuracy).
- We show that grounding does not uniformly help a strong LLM, and isolate the one mechanism that does: grounding in empirical feedback (test-and-revise) is the only regime that significantly improves in-distribution usefulness.
- We surface two cautions the field under-reports: pure elaboration (proposal writing) actively degrades idea quality and diversity, and data grounding can trade out-of-distribution robustness for in-distribution fit.
- We complement accuracy with diversity, a novelty–usefulness analysis (a task-difficulty confound that vanishes within task), and a cost/benefit accounting, and release all banks, predictions, and statistics for reproducibility.

The rest of the paper reviews related work (section 2), details the protocol and six regimes (section 3), presents results (section 4), and discusses implications and limitations (section 5).

2 Related Work

Literature-grounded ideation. A large body of work grounds idea generation in retrieved prior art. SCIMON retrieves inspirations and runs an iterative “novelty-boosting” loop that forces each idea to differ from its k nearest neighbors, raising human-judged novelty for 46–58% of ideas [Wang et al., 2024]. ResearchAgent traverses citation graphs and an entity knowledge store [Baek et al., 2025]; Chain-of-Ideas induces the developmental trend of a field before proposing [Li et al., 2025]; Nova plans and retrieves each round to keep ideas novel [Hu et al., 2024]; and MOOSE-Chem composes hypotheses from background plus deliberately non-neighboring inspirations [Yang et al., 2025]. We instantiate this family as our literature- comparison regime, but—to keep the ablation leakage-controlled and single-model—we differentiate against the model’s own accumulating idea pool rather than an external corpus. Unlike these systems, our goal is not to build a better literature-

grounded generator but to place literature comparison *on the same footing* as other grounding mechanisms.

Data-driven and empirical grounding. HYPOGENIC induces explanatory hypotheses from labeled examples and refines them with a bandit-style reward tied to predictive accuracy, beating few-shot prompting by 3.3–31.7% and even fine-tuned oracles [Zhou et al., 2024]. Literature Meets Data shows literature and data are complementary, improving OOD accuracy by 8.97% over few-shot [Liu et al., 2024]. SGA closes the loop with a differentiable simulator, where removing the simulation grounding collapses performance by orders of magnitude [Ma et al., 2024]. HYPOBENCH benchmarks these methods at scale and finds data-driven and literature+data methods dominate inference-only and zero-shot-generation baselines, especially OOD [Liu et al., 2025a]. Our induction and empirical regimes are direct, minimal instantiations of this line; our contribution is to measure them against the *same* ungrounded control and each other, and to report when data grounding *overfits* rather than transfers.

Proposal-writing and structured ideation. Several systems ground ideation by writing full structured proposals (problem $\rightarrow$ method $\rightarrow$ experiments) that are critiqued and revised [Baek et al., 2025, Si et al., 2025]. Si et al. [2025] find LLM proposals are judged more novel than expert humans but slightly less feasible. We include proposal writing as a regime and find that, when distilled into decision rules, it can *reduce* usefulness and diversity—a concrete case of an idea looking thorough while being less useful.

Evaluating ideas. Because “is this novel?” prompts are unreliable, recent work grounds the *judgment* in retrieved literature: RAG-Novelty and literature-grounded assessment beat ungrounded LLM-as-judge prompting [Lin et al., 2024, Shahid et al., 2025], and idea-level benchmarks catalog the trade-offs [Guo et al., 2024, Liu et al., 2025b]. The consistent caveat—*novelty $\neq$ value*, sharpened by the ideation–execution flip of Si et al. [2025]—motivates our choice to score grounded usefulness (downstream IND/ODD accuracy) rather than a bare novelty score, and to test the novelty–usefulness relationship directly.

Our position. Every method above grounds in a single way and beats an ungrounded prompt measured with a weaker model. We hold task, model, generation budget, and evaluation fixed and ablate the grounding *mechanisms* against one another under one protocol. To our knowledge this head-to-head comparison does not exist, and it yields a partly negative, mechanism-specific picture that the single-grounding literature could not surface.

3 Methodology

Operationalizing the question. We make “idea quality” objective and leakage-controlled. An *idea* is a *task hypothesis*: an explicit, general decision rule explaining a binary classification phenomenon (e.g., “stressed posts use more first-person pronouns and absolutist words”). “**Better**” is *grounded usefulness*: when an LLM applies a full bank of $K = 15$ hypotheses to a new instance, how often it predicts the correct label. This reuses the field-standard HYPOGENIC utility metric [Zhou et al., 2024] and lets us compare grounding mechanisms apples-to-apples.

3.1 Six ideation regimes

The independent variable is the ideation regime, with six levels: one ungrounded control and five grounding mechanisms drawn directly from the strategies named in the literature (section 2). table 1 summarizes them. Held fixed across all regimes: the generator GPT-4.1-MINI ($T=0.7$), the bank size $K=15$, and the identical evaluation procedure below. The only thing that varies is *how the ideas were grounded*. To avoid leakage, the dataset’s expert known_hypotheses are *never* shown during generation; the literature-comparison regime differentiates against the model’s own accumulating pool, not the expert set.

3.2 Shared evaluator

For every bank, a separate evaluator LLM (GPT-4.1-MINI, $T=0$) is shown the 15 hypotheses plus one instance and must apply them to answer “A” or “B”. The option order is deterministically swapped per item to neutralize position bias. Unparseable answers count as wrong (we track an

Table 1: The six ideation regimes: one ungrounded control plus five grounding mechanisms. Each generates a bank of $K=15$ task hypotheses under GPT-4.1-MINI. Generation tokens per bank (right-most) span nearly two orders of magnitude.

ID	Regime	Grounding	What it sees / does	Tok/bank
C0	UNGROUNDING (control)	none	task description only $\rightarrow$ “propose 15 novel hypotheses”	460
C1	DEDUCTION	reasoning	decompose causal mechanisms from first principles $\rightarrow$ derive rules	918
C2	INDUCTION	data	20 labeled examples $\rightarrow$ induce explanatory rules	2,897
C3	EMPIRICAL	empirical feedback	induce, then <i>test</i> the bank on fresh examples, observe errors, <i>revise</i> (2 rounds)	36,935
C4	LITCOMPARE	comparison	generate in rounds, each forced to differ from the prior pool (SCIMON-style)	1,150
C5	PROPOSAL	elaboration	write a structured proposal $\rightarrow$ distill into rules	1,588

abstain rate, which was 0.0% everywhere). Each bank is scored on 100 IND-test and 100 OOD items. Because the evaluator prompt and items are identical across regimes, differences in accuracy reflect differences in the *hypotheses*, not the inference procedure.

3.3 Datasets

We use three HYPOGENIC *real* tasks, all balanced 50/50: HEADLINE (which of two headlines gets more clicks), DECEPTIVE (fake vs. genuine hotel review), and DREADDIT (stress vs. no-stress Reddit post). Each ships a task description and a set of expert known_hypotheses, which we hold out as a novelty anchor and an approximate topline reference (never shown during generation). DECEPTIVE and DREADDIT provide an OOD split from a different source/domain; for HEADLINE we use a disjoint held-out split.

3.4 Baselines and statistics

The ungrounded control C0 is the key reference; chance/majority is 0.50 (balanced tasks), and the expert known-hypotheses bank is an approximate topline. With 3 seeds $\times$ 3 tasks we have 9 paired cells per regime. Our primary test compares each grounded regime against C0, paired across cells, using the Wilcoxon signed-rank test and a paired t -test, reporting Cohen’s d effect size and applying a Bonferroni correction over the 5 grounded-vs-control comparisons. We compute bootstrap 95% confidence intervals (10k resamples) over the pooled test items ($n=900$ per regime per split). We set $\alpha = 0.05$.

3.5 Reproducibility and cost

Seeds are fixed and every API call is disk-cached for idempotent reruns; banks, per-item predictions, and statistics are all saved. The full study (3 tasks $\times$ 6 regimes $\times$ 3 seeds, 100 IND + 100 OOD eval items) used 7.19M input and 44.8k output tokens across 11,931 calls ($\approx$ \$3), with a wall-clock of $\approx$ 14 minutes at 32-way concurrency. No GPU is required; the pipeline is API-only.

4 Results

We organize results around four questions: Does grounding help overall (section 4.1)? Which mechanism helps, and is it significant (section 4.2)? How does the answer vary by task (section 4.3)? And do novelty, diversity, and cost track usefulness (section 4.4)?

4.1 Grounding does not uniformly help

table 2 reports mean accuracy per regime, averaged over 3 tasks $\times$ 3 seeds. The empirical regime (C3) has the best IND accuracy (0.676), and the ungrounded control (C0) has the best OOD accuracy

Table 2: Aggregate accuracy by regime (mean over 3 tasks $\times$ 3 seeds). **Best per column in bold.** Grounding does not uniformly beat the ungrounded control C0: C3 (empirical) is best IND, but C0 is best OOD. Diversity is mean pairwise cosine distance within a bank; novelty distance is mean cosine distance to the held-out expert hypotheses; generation tokens per bank span $\sim 80\times$.

Regime	IND acc	IND sd	OOD acc	OOD sd	Diversity	Novelty	Gen tok/bank
C0 UNGROUNDED (control)	0.631	.079	0.760	.183	0.417	0.399	460
C1 DEDUCTION	0.608	.071	0.766	.166	0.367	0.399	918
C2 INDUCTION	0.637	.053	0.719	.113	0.409	0.431	2,897
C3 EMPIRICAL	0.676	.095	0.739	.117	0.392	0.430	36,935
C4 LITCOMPARE	0.601	.032	0.721	.175	0.396	0.415	1,150
C5 PROPOSAL	0.574	.070	0.729	.183	0.308	0.415	1,588

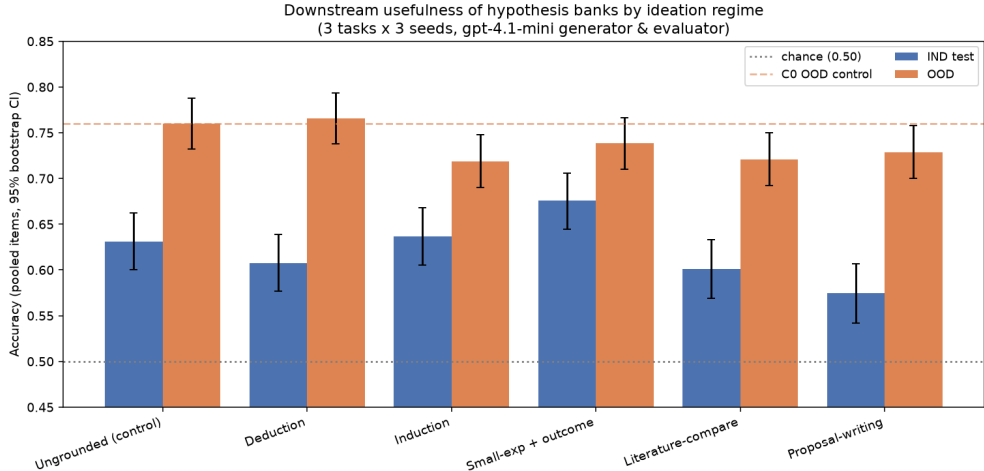


Figure 1: In-distribution (IND) and out-of-distribution (OOD) accuracy by regime, with bootstrap 95% confidence intervals. The empirical regime (C3) leads IND; the ungrounded control (C0) leads OOD. The chance line is at 0.50. Mean differences between most grounded regimes and the control are small relative to the confidence intervals.

(0.760). Pooling the five grounded regimes against C0 (our H1 test), grounding is statistically indistinguishable from—and slightly below—the control: IND 0.619 vs. 0.631 ($d = -0.32$, $p=0.37$) and OOD 0.735 vs. 0.760 ($d = -0.41$, $p=0.25$). The blanket claim “grounding makes a strong LLM’s ideas better” is *not* supported.

4.2 The mechanism matters: empirical grounding helps, proposal writing hurts

table 3 gives the paired comparison of each grounded regime against the control. Two regimes move the needle, in opposite directions. EMPIRICAL (C3) is the only regime that *significantly improves* IND accuracy ($+0.044$, $d=1.09$, Wilcoxon $p=0.020$); it is the one mechanism that changes its ideas in response to evidence—it tests the bank, sees concrete mistakes, and revises. PROPOSAL (C5) *significantly hurts* IND accuracy (-0.057 , $d = -1.56$, $p=0.008$), surviving the Bonferroni correction, and it has by far the lowest diversity (0.308 in table 2). The remaining regimes (deduction, induction, literature comparison) are within noise of the control. No grounded regime significantly beats the control OOD. This supports H2 only partially—empirical grounding leads on IND, but the full predicted ordering does not hold—and refutes H3: the grounding advantage is *not* larger OOD, and is often reversed (figure 2).

4.3 Three tasks, three stories: heterogeneity and overfitting

The aggregate masks sharp per-task heterogeneity (table 4, figure 3). On DREADDIT, empirical grounding (C3) wins clearly on *both* IND (0.78 vs. 0.72) and OOD (0.81 vs. 0.75)—here grounding genuinely helps. On DECEPTIVE, the ungrounded model transfers near-perfectly OOD (0.97) using

Table 3: Paired significance of each grounded regime vs. the ungrounded control C0 (9 paired task×seed cells; Bonferroni $m=5$). **Bold** marks significant effects ($p < 0.05$). Empirical grounding (C3) is the only regime that significantly improves IND accuracy; proposal writing (C5) significantly hurts and survives Bonferroni correction. No grounded regime significantly beats C0 OOD.

Split	Regime	Gain vs. C0	Cohen’s d	Wilcoxon p	$t p$ (Bonf.)
IND	C1 DEDUCTION	−0.023	−0.51	0.195	0.826
IND	C2 INDUCTION	+0.006	+0.11	0.785	1.000
IND	C3 EMPIRICAL	+0.044	+1.09	0.020	0.058
IND	C4 LITCOMPARE	−0.030	−0.46	0.195	1.000
IND	C5 PROPOSAL	−0.057	−1.56	0.008	0.008
OOD	C1 DEDUCTION	+0.006	+0.14	0.719	1.000
OOD	C2 INDUCTION	−0.041	−0.29	0.652	1.000
OOD	C3 EMPIRICAL	−0.021	−0.19	0.742	1.000
OOD	C4 LITCOMPARE	−0.039	−0.54	0.102	0.715
OOD	C5 PROPOSAL	−0.031	−0.47	0.234	0.988

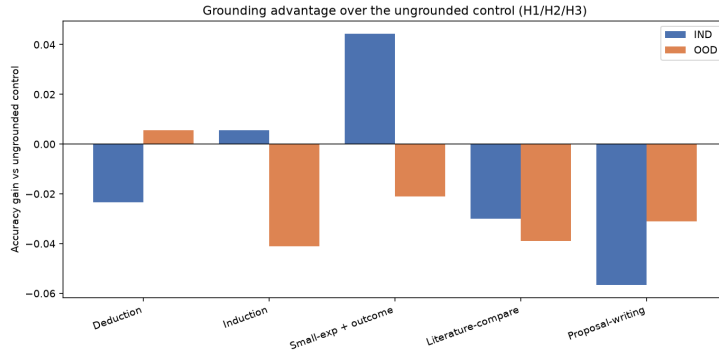


Figure 2: Grounding advantage (regime accuracy minus control C0) on IND and OOD. Only empirical grounding (C3) shows a clear positive IND advantage; proposal writing (C5) shows a clear negative one. Contrary to H3, advantages are not larger OOD—the OOD bars are mostly negative.

general priors, while data grounding (C2/C3) *raises* IND but *crashes* OOD to 0.78–0.82: the learned rules encode dataset-specific surface cues (e.g., spatial/Chicago-hotel vocabulary) that do not transfer, whereas the ungrounded prior captures general “fake review” structure that does. This is the paper’s most important nuance: *data grounding can overfit*, trading OOD robustness for IND fit—a caution prior work, which emphasizes data grounding’s OOD *wins*, under-reports. On HEADLINE, all regimes hover near chance (0.50–0.61), including the expert hypotheses (0.56/0.58); there is little extractable signal, so the task contributes mostly noise to the aggregate.

4.4 Cost, novelty, and diversity do not buy usefulness

Cost/benefit. C3’s IND gain is expensive: it costs $\sim 80\times$ the generation tokens of the control (36.9k vs. 0.46k tokens/bank), driven by its test-and-revise loop (figure 4a). The cheaper grounded regimes (C1, C4, C5 at 0.9k–1.6k tokens) do *not* beat the control. Grounding’s one reliable win comes only at the top of the cost curve.

Novelty $\neq$ usefulness. Across the 18 task×regime cells, Spearman(novelty, OOD) = +0.835 ($p < 0.001$)—but this is a task-difficulty confound (a Simpson’s paradox): easy, high-accuracy tasks also have ideas farther from their expert anchors. *Within* a task the relationship vanishes (mean $\rho \approx -0.10$; DECEPTIVE $\rho = -0.75$), and diversity→usefulness is weak within task (mean $\rho \approx +0.33$, n.s.). Novelty and diversity are therefore not reliable proxies for usefulness (figure 4b), consistent with the literature’s caveat and confirming H4 once the task confound is removed.

Table 4: Per-task accuracy (IND / OOD, mean over seeds) for selected regimes. **Bold** marks the best IND/OOD on a task among the shown regimes. Three distinct stories: grounding helps on DREADDIT, overfits on DECEPTIVE (ungrounded OOD 0.97 vs. grounded ≤ 0.82), and finds no signal on HEADLINE.

Regime	HEADLINE	DECEPTIVE	DREADDIT
C0 UNGROUNDED	0.55/0.56	0.62/ 0.97	0.72/0.75
C2 INDUCTION	0.58/0.61	0.65/0.78	0.68/0.77
C3 EMPIRICAL	0.56/0.59	0.69 /0.82	0.78 / 0.81
C5 PROPOSAL	0.50/0.57	0.58/0.97	0.64/0.65

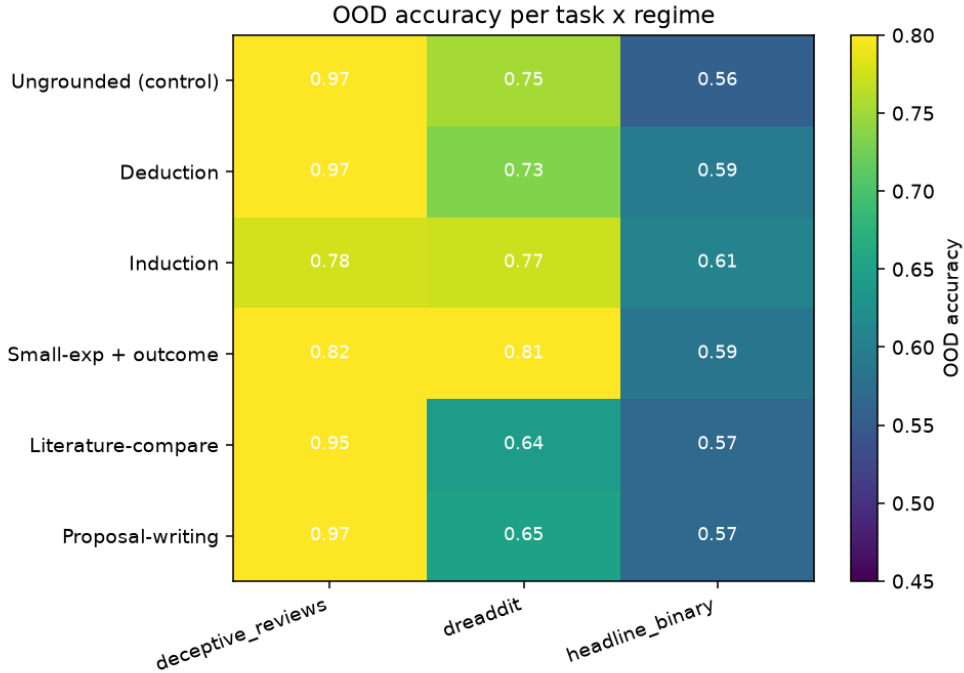


Figure 3: Per-task accuracy grid across all six regimes (IND and OOD). The heterogeneity is stark: DREADDIT rewards grounding, DECEPTIVE shows the ungrounded control’s near-perfect OOD transfer collapsing under data grounding, and HEADLINE sits near chance throughout.

5 Discussion

Why ungrounded is so strong now. The control’s banks already recite the *right* expert-level features. For DREADDIT, GPT-4.1-MINI ungrounded produces textbook stress markers—first-person pronouns, absolutist words, sleep disturbance, inability to cope—without ever seeing data. A strong model’s parametric prior is itself a form of grounding, acquired in pretraining. This is the crux of our partly negative result: the 2023-era finding “grounding beats ungrounded prompting” was measured against much weaker baselines; with a 2025 model the ungrounded control is a strong, often-transferable starting point, which shrinks grounding’s headroom.

Why empirical grounding is the exception. EMPIRICAL (C3) is the only regime that changes its ideas in response to evidence: it tests the bank, observes concrete mistakes, and revises, producing richer rules (e.g., “intrusive thoughts / trauma / self-blame with ongoing distress”). This closed loop is exactly the “small experiments + outcome imagination” grounding the hypothesis names, and it is the one mechanism that reliably lifts IND accuracy ($d=1.09$). The catch is twofold: its IND gain does not transfer OOD, and it is by far the most expensive regime ($\sim 80\times$ tokens).

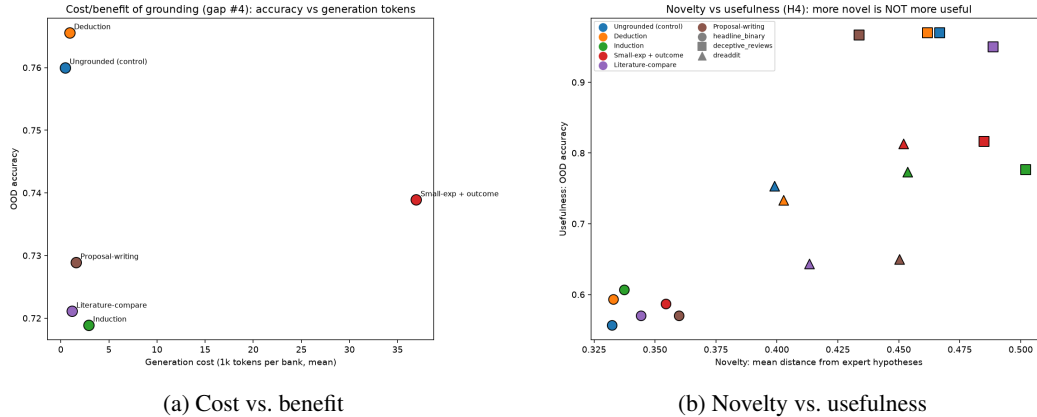


Figure 4: (a) (Left) Generation tokens per bank vs. IND accuracy: empirical grounding (C3) buys its gain at $\sim 80\times$ the control’s token cost, while cheaper grounded regimes do not beat the control. (b) (Right) Bank novelty (distance from expert hypotheses) vs. OOD usefulness: the strong pooled correlation is a task-difficulty confound that vanishes within task, so novelty does not track usefulness.

Why proposal writing backfires. Distilling a prose proposal into rules yields rigid, symmetric “if high $X \rightarrow A$ / if low $X \rightarrow B$ ” statements with the lowest diversity of any regime. The elaboration adds words, not discriminative power, and the brittle thresholds misfire. This is a concrete instance of LLM ideation that looks thorough while being measurably less useful—a warning against equating structured elaboration with idea quality.

The overfitting result. On DECEPTIVE, data grounding helped IND but *hurt* OOD, because the learned rules encode dataset-specific surface cues that do not transfer, whereas the ungrounded prior captures general structure that does. Prior work emphasizes data grounding’s OOD *wins*; under one protocol with a strong model we find it can *trade* OOD robustness for IND fit. (Here DECEPTIVE’s OOD split is in fact easier than its IND split for general priors, which amplifies the effect; the within-split grounded-vs-ungrounded comparison still holds.) The practical lesson is that “ground in data” should come with an OOD check, not an assumption of transfer.

Verdict on the hypotheses. The hypothesis “grounding makes LLM ideas better” is *refuted as a blanket claim* but *supported for one specific mechanism*. H1 (grounding helps overall) is not supported. H2 (empirical grounding strongest) is partially supported: empirical grounding leads IND, but the full predicted ordering does not hold. H3 (grounding helps most OOD) is not supported—often the reverse. H4 (novelty $\neq$ usefulness) is confirmed once the task-difficulty confound is removed. The headline takeaway: “ground your ideation” is too coarse a prescription for today’s strong LLMs; *grounding in empirical feedback* is the one mechanism that reliably adds value, pure elaboration harms, and data grounding’s benefit is conditional.

Limitations. (1) *Tasks and construct.* We use three binary text-classification tasks (one near chance), and operationalize “idea quality” as downstream label accuracy. This captures grounded usefulness but not open-ended scientific creativity; idea-level benchmarks were out of scope. (2) *Single model family.* Generator and evaluator are both GPT-4.1-MINI. A stronger ungrounded baseline would likely shrink grounding’s value further, while a weaker model might restore the classic grounding wins; cross-model replication is the key next step. (3) *Power.* We have 9 paired cells per comparison (and only 6 points per task for the within-task novelty analysis); effect *directions* are clear, but some non-significant results may be power-limited, and HEADLINE contributes mostly noise. (4) *One instantiation per regime.* Each grounding mechanism is one reasonable design; stronger literature grounding (real retrieval over papers) or longer empirical loops could change the standings. (5) *Evaluator ceiling.* A single shared evaluator is fair across regimes but caps the measurable quality of any hypothesis.

Broader implications. For practitioners, the message is to spend grounding effort where it pays: a cheap ungrounded prompt of a capable model is a strong default, empirical test-and-revise is worth its cost when in-distribution accuracy is the goal and budget allows, and structured elaboration should be treated with suspicion. For researchers, single-grounding comparisons against weak

baselines can overstate grounding’s value; mechanism-level, single-protocol ablations with strong controls give a more honest picture.

6 Conclusion

We presented the first single-protocol, head-to-head ablation of six LLM ideation regimes—one ungrounded control and five grounding mechanisms—holding the model, task suite, generation budget, and evaluation procedure fixed and scoring each by objective downstream utility. Our answer to the motivating question is nuanced and partly negative: under a fixed, objective protocol, *grounding does not reliably make a strong LLM’s ideas better*. Pooled across mechanisms, grounded banks are statistically indistinguishable from a strong ungrounded control.

Key findings. The mechanism is what matters. Grounding in empirical feedback (test-and-revise) is the only regime that significantly improves in-distribution usefulness (+4.4 points, $d=1.09$), but it costs $\sim 80\times$ the tokens and its gains do not transfer out-of-distribution. Pure elaboration (proposal writing) significantly *hurts* and collapses idea diversity. Data grounding can *overfit*, trading out-of-distribution robustness for in-distribution fit. And novelty and diversity do not track usefulness within a task. Ungrounded prompting of a capable model is a surprisingly hard baseline to beat.

Future work. Four directions follow directly. (1) A cross-model sweep from weaker to frontier generators, to map where grounding’s value lives as the ungrounded baseline strengthens. (2) Regularized or OOD-aware empirical grounding that keeps the in-distribution gain without overfitting to surface cues. (3) Idea-level domains (e.g., rediscovery benchmarks) to test whether these conclusions hold for open-ended ideation rather than classification rules. (4) Hybrid “prior + minimal empirical check” regimes that aim to capture most of empirical grounding’s gain at a fraction of its cost. We release all banks, per-item predictions, and statistics to support these next steps.

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